Who Gains from Generative AI Model Releases?

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Abstract

Stock-market reactions to generative AI releases concentrate in two ecosystem positions. We study 124 day-verified releases from 2022 to 2026 using 45 securities in the Nasdaq-100 Technology Sector Index, coded by their relationships to 25 model developers. Hardware suppliers and rival model developers earn 3.48 and 3.39 percentage points over twenty trading days; no other position reaches the 5% significance level. Both premia emerge in 2025–2026, and the increase for hardware exceeds that for competitors by about 5 percentage points. Capability scores carry no positive average premium. A one-standard-deviation increase in capability widens the hardware premium by about 2 percentage points. The results show how model-release reactions vary across ecosystem position, market development, and model capability.

Keywords: Generative AI, Event study, Ecosystem position, Model capability, Technology stocks

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1. Introduction

A generative AI model release reaches several markets at once. It introduces a product for its developer, signals the growth of an emerging industry to rival developers, and can shift expected demand for semiconductors, servers, and networking equipment. Firms commonly grouped under "AI exposure" therefore occupy economically distinct positions around the same announcement. A release can validate the commercial prospects of the model industry, raising the value of competing developers. It can also change expectations about demand for the physical inputs used to train and serve models, raising the value of hardware suppliers.

We measure these position-specific reactions using 124 model releases from 2022 to 2026 and 45 securities in the Nasdaq-100 Technology Sector Index (NDXT). Release dates are verified against first-party sources. Two coders independently classify each firm–developer pair into eight ecosystem positions. The outcome is the firm's cumulative abnormal return over the twenty trading days following a release.

Release returns concentrate in hardware suppliers and rival model developers. Hardware suppliers earn 3.48 pp and competitors earn 3.39 pp relative to firms outside each position. Both coefficients are significant at 1% with event-level CR2 inference and at 5% with event-and-firm clustering. Cloud platforms, integrators, deployers, enablers, investors, and listed owners do not reach the 5% level under either inference method.

The concentration changes sharply after 2025. Hardware and competitor coefficients are small and statistically insignificant during 2022–2024, then rise to 7.18 pp and 4.61 pp during 2025–2026. The hardware increase exceeds the competitor increase by 4.86 pp under the market model and 5.03 pp under the Fama–French three-factor model. The difference is significant under event-level and two-way inference.

Model capability further differentiates the hardware response. The Artificial Analysis intelligence index has no positive pooled return slope. Its interaction with hardware is positive under both return benchmarks. A one-standard-deviation increase in capability widens the hardware premium by approximately 2 pp.

The evidence extends asset-pricing work that measures firms' broad exposure to AI (Eisfeldt et al., 2023; Babina et al., 2024). Position coding separates suppliers, developers, intermediaries, and users inside the technology sector. The positive competitor response also connects generative AI releases to the contagion component of intra-industry news (Lang and Stulz, 1992; Chen et al., 2005). Taken together, the estimates show two distinct return profiles within the technology sector: a competitor premium shared by rival developers and a hardware premium that grows after 2025 and with model capability. The results document position-level variation in how technology stocks respond to model releases.

2. Data and Empirical Design

The event pipeline begins with public model chronologies and benchmark records. We consolidate same-day releases from the same model family and verify dates using vendor announcements, release notes, repositories, and archived first captures. Day-level evidence is available for 125 events. One identity-inconsistent event is excluded, leaving 124 releases: 3 in 2022, 8 in

2023, 45 in 2024, 59 in 2025, and 9 in 2026. Eighty-seven language-model events have a comparable Artificial Analysis intelligence score.

The firm universe contains the 45 securities listed in the NDXT constituent file dated May 1, 2026. They represent 44 issuers because Alphabet has two share classes. Two independent coders classify 1,125 firm–developer pairs, covering 45 securities and 25 model developers. The eight indicators are hardware supplier, cloud platform, integrator, deployer, downstream enabler, competitor, investor, and listed owner. Coding is multi-label. Cell-level agreement is 98.5%, and all disagreements are adjudicated under a fixed codebook.

Hardware suppliers earn primary revenue from semiconductors, servers, networking, storage, and related physical inputs. Cloud platforms rent computing capacity and host third-party models. Integrators embed foundation models in their own software, while deployers use AI inside a broader business. Downstream enablers implement AI systems for clients. Competitors develop models in the same modality as the released model. Investor and owner indicators capture financial ties and listed publishers. A firm can hold several positions around the same release; each regression includes all eight indicators jointly.

Daily adjusted returns run from January 2021 through April 2026. The market model uses the S&P 500 and an estimation window of $[-200, -11]$. The second benchmark is the Fama–French three-factor model (Fama and French, 1993). The dependent variable is $\text{CAR}[0, 20]$. Regressions include log assets, book-to-market, momentum, volatility, and year fixed effects. The main sample has 5,441 firm-event observations. We estimate

$$CAR_{ie} = \alpha + \sum_k \beta_k Position_{iek} + \gamma' Z_{ie} + \lambda_y + \varepsilon_{ie}. \tag{1}$$

The time specification adds each position's interaction with an indicator for releases in 2025–2026. The capability specification centers the intelligence index and interacts it with the hardware indicator. Tables report exact p-values from event-level CR2 and event-and-firm two-way clustering; stars use event-level CR2 inference (Cameron et al., 2011).

The coefficients compare firms holding a given position with the remaining technology firms around the same set of releases, conditional on the other positions and firm characteristics. The post-2025 interactions compare that cross-sectional pricing rule across the two periods. The final contrast tests whether the change in the hardware coefficient equals the change in the competitor coefficient.

3. Results

3.1. Position-level returns

Table 1 shows a sharp cross-sectional concentration. Hardware and competitor coefficients are similar in magnitude and statistically indistinguishable. Under the market model, their difference is 0.09 pp ($p = 0.92$). The other six position coefficients do not reach 5% under either inference method.

The hardware coefficient exceeds the average of the four downstream positions by 2.53 pp under the market model and 2.66 pp under the three-factor model. Both contrasts are significant at 1% under event-level and two-way inference (Online Appendix Table A3).

Table 1: Ecosystem position and twenty-day abnormal returns

	Market model			Three-factor model		
Position	Coef.	Event p	Two-way p	Coef.	Event p	Two-way p
Hardware supplier	3.476***	0.002	0.015	3.428***	0.002	0.016
Cloud platform	1.050	0.121	0.246	1.171*	0.079	0.235
Downstream integrator	1.315*	0.078	0.129	1.185	0.119	0.164
Downstream deployer	1.261	0.137	0.183	1.266	0.140	0.192
Downstream enabler	0.251	0.802	0.781	−0.133	0.894	0.887
Competitor	3.385***	< 0.001	0.022	3.371***	< 0.001	0.037
Investor	0.480	0.691	0.627	1.048	0.362	0.291
Listed owner	3.181*	0.070	0.113	3.229*	0.067	0.123
Observations		5,441			5,441	
Events		124			124	

Notes: Coefficients are percentage points. Regressions include all eight position indicators, firm controls, and year fixed effects. Event p uses event-level CR2 inference. Two-way p clusters by event and firm. Event-CR2 stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The SOXX benchmark sharpens the hardware result. After absorbing broad semiconductor-sector returns, hardware suppliers earn 2.90 pp ($p = 0.003$), while competitors earn 3.18 pp ($p < 0.001$). The hardware coefficient therefore extends beyond a contemporaneous rise in semiconductor stocks. The 2.53–2.66 pp advantage over downstream firms shows a stronger release response among hardware suppliers than among downstream firms within the technology sector.

The competitor coefficient carries a different economic meaning. A rival developer can lose market share to a new release, yet its value can rise if the release conveys positive news about the size or adoption of the model market as a whole. The positive 3.39 pp estimate is consistent with this

Table 2: Position premia before and after 2025

	Market model			Three-factor model		
Estimate	Coef.	Event p	Two-way p	Coef.	Event p	Two-way p
Hardware, 2022–2024	-1.285	0.463	0.431	-1.710	0.317	0.316
Competitor, 2022–2024	1.001	0.427	0.524	0.685	0.593	0.680
Hardware, 2025–2026	7.185^{***}	< 0.001	< 0.001	7.439^{***}	< 0.001	< 0.001
Competitor, 2025–2026	4.607^{***}	< 0.001	0.003	4.802^{***}	< 0.001	0.004
Post-2025 change, hardware	8.469^{***}	< 0.001	0.005	9.149^{***}	< 0.001	0.004
Post-2025 change, competitor	3.606^{**}	0.018	0.064	4.116^{***}	0.009	0.035
Hardware change minus competitor change	4.863^{***}	0.001	0.021	5.032^{***}	0.001	0.022

Notes: Coefficients are percentage points from a pooled regression containing all position indicators and their interactions with a 2025–2026 indicator. Controls and inference follow Table 1. Event-CR2 stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

industry-level channel dominating competitive displacement over the sample period. Its statistical equality with the hardware estimate makes the time comparison central to distinguishing the two return profiles.

3.2. Time variation

Table 2 traces the two premia across time. During 2022–2024, the hardware coefficient is -1.28 pp and the competitor coefficient is 1.00 pp under the market model; both are statistically insignificant. During 2025–2026, they rise to 7.18 pp and 4.61 pp, with $p < 0.01$ under both inference methods.

Both positions strengthen after 2025, but the hardware increase is 4.86–5.03 pp larger. Hardware therefore has the stronger post-2025 response.

The horizon profile in Figure 1 shows how the late-period premia accumulate. In 2025–2026, hardware and competitor coefficients are small on the release date and rise over the following twenty trading days. Hardware reaches 7.7 pp in the window-by-window specification, while the competitor coefficient reaches 5.6 pp. The growing distance between the two lines mirrors the formal difference-in-changes test.

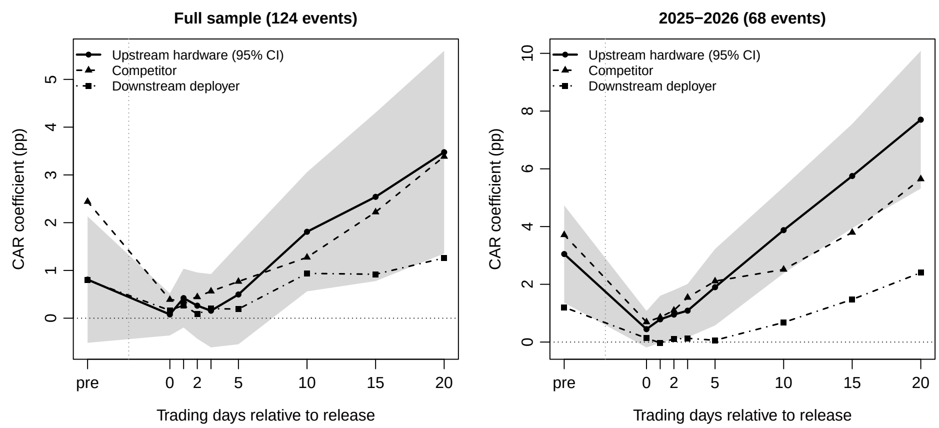

Figure 1: Cumulative abnormal returns by ecosystem position. Coefficients come from market-model regressions with firm controls and year fixed effects. The shaded area is the 95% event-clustered CR2 confidence interval for hardware. The pre-event point is CAR$[-10, -2]$; the remaining points cumulate from the release date.

3.3. Capability and the hardware premium

The pooled capability slope is -0.059 pp per index point under the market model and -0.080 pp under the three-factor model (Table 3). Capability therefore carries no positive average premium.

The interaction is 0.154 pp per index point under the market model and 0.162 pp under the three-factor model. The intelligence index has a standard deviation of 12.7 points, so a one-standard-deviation increase widens the hardware premium by 1.95–2.06 pp. Capability therefore shifts the cross-sectional allocation of release returns toward hardware suppliers.

Table 3: Model capability and the hardware premium

	Market model			Three-factor model		
Term	Coef.	Event p	Two-way p	Coef.	Event p	Two-way p
Capability, pooled	-0.059^{*}	0.071	0.082	-0.080^{**}	0.030	0.030
Hardware at mean capability	1.853^{**}	0.046	0.068	1.950^{**}	0.038	0.056
Capability × hardware	0.154^{**}	0.045	0.051	0.162^{**}	0.034	0.039
Observations		3,842			3,842	
Events		87			87	

Notes: Coefficients are percentage points. Capability is the Artificial Analysis intelligence index. The interaction regressions center the index before multiplying it by the hardware indicator. All specifications include firm controls and year fixed effects. Event-CR2 stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The interaction links technical performance to cross-sectional variation in the hardware premium. A higher benchmark score is associated with a larger return difference between hardware suppliers and other firms. One interpretation is stronger expected demand for computing inputs. The same position also shows the largest post-2025 increase.

4. Conclusion

Generative AI model releases are priced unevenly across the technology sector. Hardware suppliers and rival model developers show positive twenty-day abnormal returns. Both premia emerge after 2025, with an additional hardware increase of about 5 pp. Model capability widens the hardware premium by about 2 pp per standard deviation. Ecosystem position, market development, and model capability jointly shape the cross-section of release-related stock returns.

The results highlight the value of distinguishing ecosystem positions when studying AI-related news. Hardware and competitor premia are similar over the full period, but their time profiles diverge after 2025, and only the hardware premium rises with model capability.

Data availability

The event pipeline, coding decisions, estimation code, and complete robustness results are available from the authors. The online appendix reports the coding protocol, alternative benchmarks, dependence adjustments, non-overlapping-event samples, and composition tests.

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